Galaxy Simulator using N-Body Gravitational Physics

Abhishek
Department of Physics (Student)
India
Email: (your email)

Abstract

This paper presents the development of a real-time galaxy simulation using an N-body gravitational model implemented in Python. The system simulates the interaction of multiple stars under Newtonian gravity, resulting in the emergence of spiral galaxy-like structures. A central massive object is introduced to stabilize orbital motion and represent a galactic core. The simulation demonstrates key astrophysical concepts such as orbital mechanics, angular momentum conservation, and density variations within galactic disks. The project serves as an introduction to computational astrophysics and numerical simulation techniques.

Keywords

N-body Simulation, Galaxy Formation, Computational Astrophysics, Orbital Mechanics, Gravitational Systems

---

I. Introduction

Galaxies are large-scale gravitational systems consisting of billions of stars interacting over cosmic timescales. Understanding their structure and evolution requires computational approaches due to the complexity of many-body interactions.

N-body simulations provide a numerical method to approximate such systems by modeling individual particles under gravitational forces. This project aims to simulate galaxy formation and dynamics using a simplified computational model.

The objectives of this work include:

- Modeling gravitational interaction between multiple stars
- Simulating orbital motion around a central mass
- Visualizing spiral galaxy structures

II. Theoretical Background

A. Newtonian Gravitation

The gravitational force between two masses is given by:

$F = G m_1 m_2 / r^2$

where G is the gravitational constant, m_1 and m_2 are masses, and r is the distance between them.

B. N-Body Systems

In an N-body system, each particle interacts with every other particle:

$F_i = \Sigma (G m_i m_j / r^2)$

This leads to complex motion requiring numerical approximation.

C. Orbital Motion

Stable circular velocity is approximated by:

$v = \sqrt{(GM / r)}$

This ensures stars maintain orbit around the galactic center.

D. Spiral Structure Formation

Spiral arms are explained using density wave theory, where stars move through regions of higher density, creating visible spiral patterns.

III. Methodology

A. System Design

The simulation consists of:

- Stars represented as particles
- A central massive object (black hole)
- Real-time updates using discrete time steps

B. Initialization

Stars are initialized in a disk distribution using:

- Random radial distance

- Angular positioning
- Circular velocity with perturbations

C. Force Computation

Gravitational forces are computed:

- Between stars (reduced interaction for stability)
- Between each star and central mass

A softening factor is used to prevent numerical instability.

D. Numerical Stability

To ensure realistic simulation:

- Velocity clamping limits extreme motion
- Distance softening prevents singularities
- Reduced inter-particle force avoids chaotic behavior

E. Visualization

The system uses Pygame to render:

- Star motion in real-time
- Trail effects to visualize orbits
- Brightness variation based on radial distance

IV. Results

The simulation produces:

- Spiral galaxy-like structures
- Stable orbital motion
- Central density concentration
- Smooth rotational behavior

Figure 1: Structure of a Spiral Galaxy
:- Structure of a Spiral Galaxy showing the galactic bulge, disk, and spiral arms.

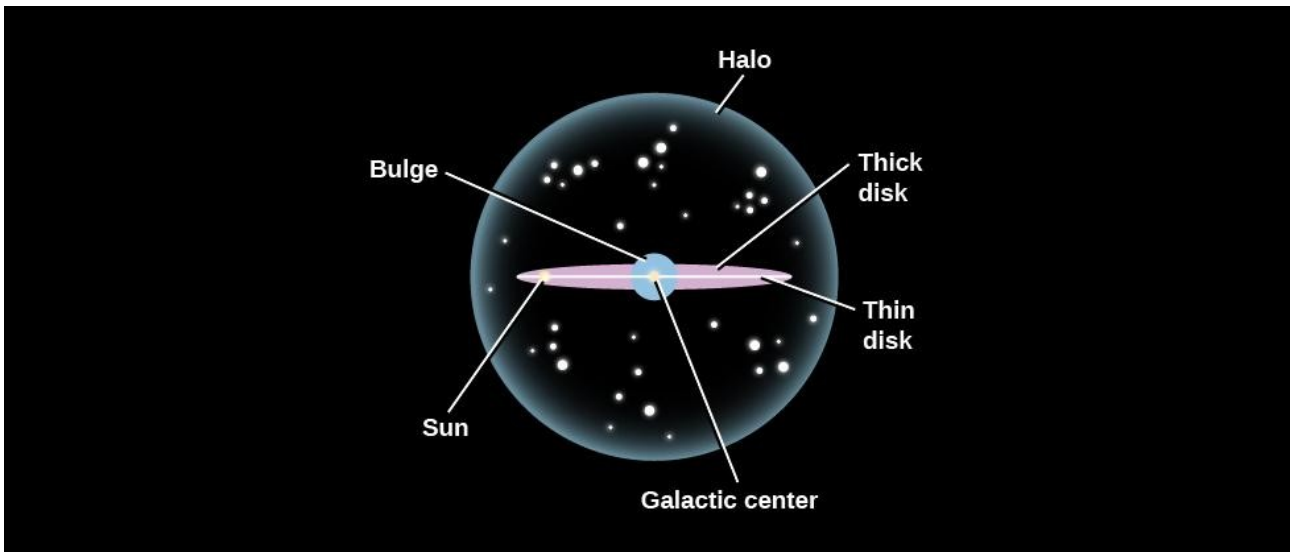
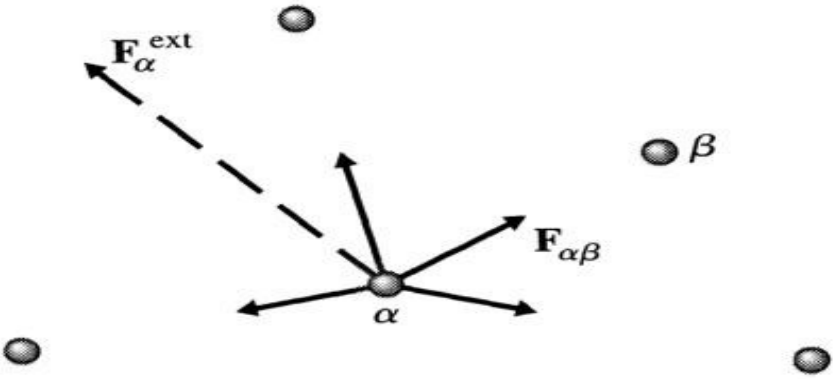
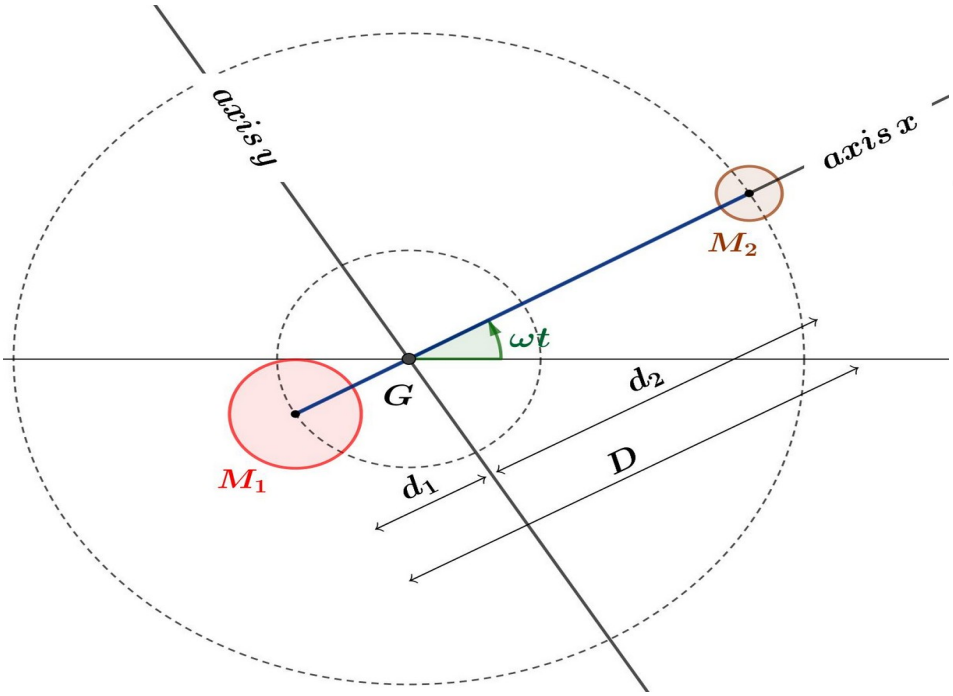


Figure 2: N-body Interaction Model



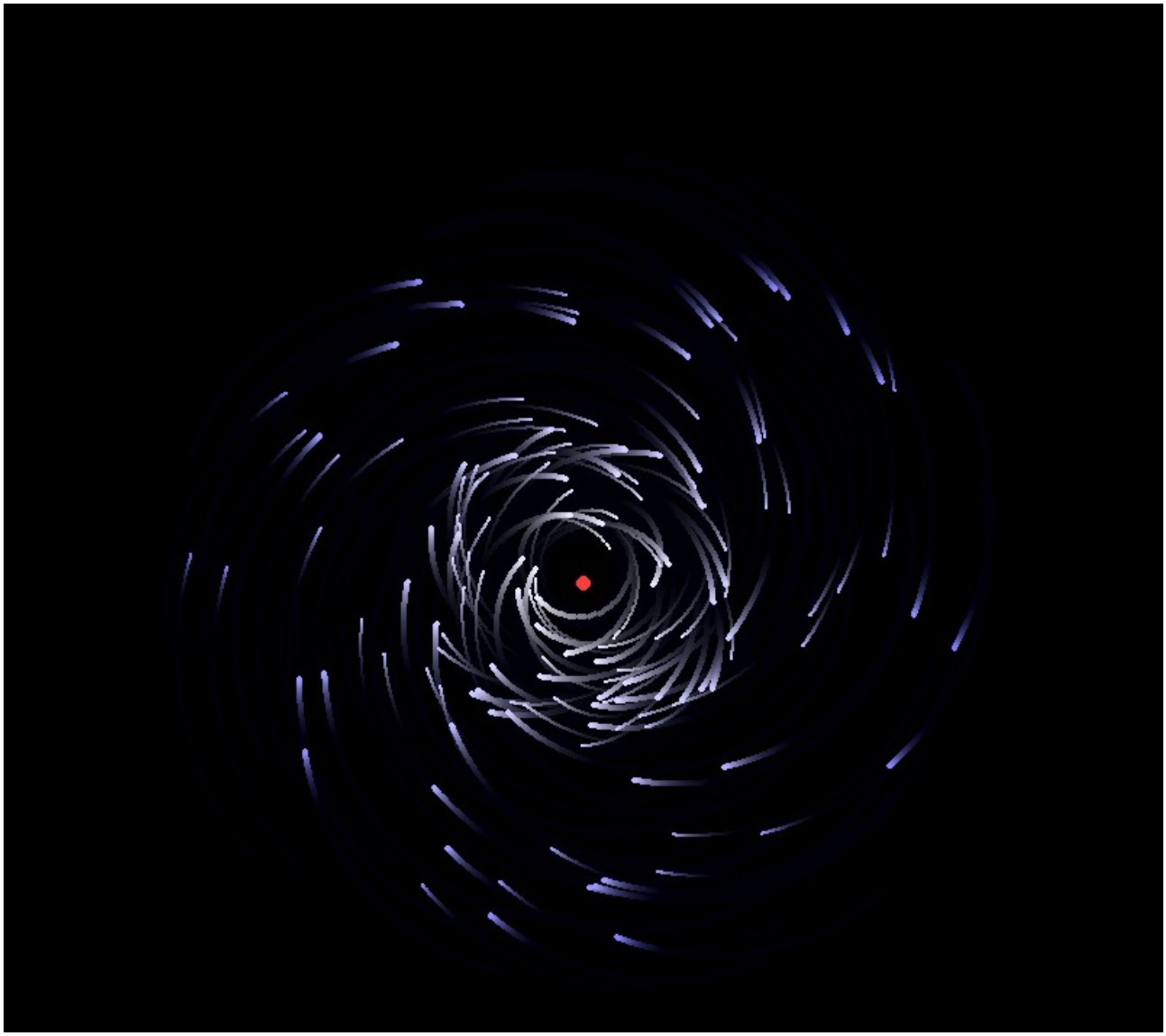
:- Representation of N-body simulation where each particle interacts gravitationally with others.

Figure 3: Orbital Motion around Central Mass



:- Orbitalmotion of stars around a central massive object (blackhole).

Figure 4: Simulation Output



:- Output of the developed galaxy simulation showing spiral structure and central density.

V. Discussion

The simulation successfully captures key characteristics of galactic systems. The emergence of spiral structures and stable orbits demonstrates the effectiveness of simplified gravitational modeling.

However, limitations include:

- Limited number of particles
- Absence of dark matter modeling
- No relativistic effects

VI. Future Work

Future improvements include:

- Dark matter halo simulation
- Galaxy collision modeling
- 3D visualization
- Barnes-Hut algorithm optimization
- Integration with machine learning techniques

VII. Conclusion

This project demonstrates how fundamental physical laws can be used to simulate complex astrophysical systems. The galaxy simulator provides a visual and computational understanding of gravitational dynamics and serves as a foundation for further research in computational astrophysics.

References

[1] I. Newton, *Philosophiæ Naturalis Principia Mathematica*
[2] J. Binney and S. Tremaine, *Galactic Dynamics*
[3] A. Toomre, “On the gravitational stability of a disk of stars,” 1964
[4] V. Springel, “The cosmological simulation code GADGET,” 2005